



Founders, ParkoBot

Designing a device for India's chaotic conditions was not an easy task. Early prototypes failed against dust, waterlogging, and heavy misuse, forcing multiple redesigns. The final product is an IP65-rated, cold-rolled steel barrier built to withstand loads up to five tons. Power management was another challenge. Since many sites lacked reliable power, ParkoBot engineered the smart barrier to run from six months to two years on batteries, depending on usage. It currently runs on Duracell dry cell batteries, with lithium models planned for the future.

On the electronics front, each barrier houses a custom in-house PCB and proximity sensors to ensure reliability. Bluetooth enables local communication, while cloud connectivity manages booking and analytics. For underground basements, where signals are weakened, ParkoBot has partnered with telecom providers to ensure stable network performance. Technology alone was not enough. ParkoBot had to convince property owners, most of whom were unfamiliar with monetising parking. The company rolled out incentives such as pre-booking offers, hardware subsidies, and even perks like free electric car shades.

Keeping engineering in-house proved critical. From PCB design and sensor calibration to prototyping and software, ParkoBot retained full control while outsourcing large-scale manufacturing to partners in Kolkata, Chennai, Taiwan, and China. Devices were rigorously tested in flooded basements, dusty lots, and high-traffic sites to fine-tune durability and reliability. The business model is equally innovative: hardware plus revenue share.

Owners purchase the barrier, while ParkoBot earns a commission on each booking. For drivers, flexible options include pay-per-use or subscription packages.

Today, ParkoBot manages over 2500 parking spaces and facilitates more than 20,000 bookings every month across India. Its multilingual app, available in multiple global languages, supports adoption across diverse communities. Community-specific models include:

- Residential complexes→guest-only sharing
- Offices→hourly slot conversion
- Malls→premium spots near elevators, often reimbursed by merchants
- Urban hubs→digital allocation in crowded zones

By reducing time spent hunting for parking, ParkoBot helps save fuel, cut emissions, and decongest roads. Integrated EV charging turns idle spaces into green energy assets. ParkoBot is now expanding into tier-two Indian cities and branching into two-wheeler, truck, and bus parking. International pilots in France, Dubai, and Qatar signal their global ambitions. Backed by fresh seed funding, the company is strengthening infrastructure, forging partnerships with banks, telecoms, and property associations, and adding new features:

- Priority parking
- Mall customer reimbursements
- Helmet lockers for two-wheelers
- AI-driven parking management

ParkoBot is more than just hardware. It is a resilient IoT ecosystem designed to withstand dust, flooding, misuse, and unreliable power. Its vision is to transform every slot into a multi-use urban asset, from EV charging hubs to logistics nodes and micro-retail spaces.

By merging electronics engineering, IoT, and smart incentives, ParkoBot demonstrates how technology can reshape urban mobility, one slot at a time. **EFY**

—Akanksha Sondhi Gaur

embedded electronics and IoT-driven parking management solution. Unlike conventional boom barriers placed at entry points, ParkoBot operates directly at the slot level. Each device is a rugged, waterproof barrier that prevents unauthorised parking, connects securely to the cloud, and enables:

- Time-based booking and secure payments
- AI-driven dynamic pricing
- Real-time visibility of available slots

Property owners, whether in offices, malls, residential complexes, or homes, can install the barrier in their parking slots. Whenever the space is free, it can be listed automatically on the ParkoBot app or set for manual approval by the owner. Drivers gain stress-free, bookable parking; owners earn a share of revenue, with some slots generating up to ₹300,000 annually. The system also integrates with EV charging, transforming idle office and mall spaces into overnight charging hubs, particularly attractive for fleet operators. In this way, ParkoBot simultaneously addresses congestion and accelerates India's green mobility transition.